

Beneficial infections of the enterovirus genus in patients with liver cancer

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ABSTRACT

Background Hepatocellular carcinoma (HCC) is a significant global cancer burden, with rising incidence and lacking a unified prevention strategy due to complex aetiologies. Viral exposures may shape host immunity via specific reactive viral antigens that could induce immune responses against hepatocarcinogenesis.

Objective We aimed to characterise viral exposure differences between HCC patients and healthy individuals and identify potentially protective viral antigens against HCC.

Design We profiled pan-viral serological antibody repertoires using a microbial phage library among 2647 study subjects and examined the biological activities of selective viral antigens on blood-derived immune cells from both healthy individuals and HCC patients.

Result We identified 153 viral antigens with a significantly reduced serological response in HCC patients compared with healthy individuals. We also observed that a higher serological response to 153 viral antigens is associated with better clinical outcomes of patients with chronic liver diseases and HCC. These findings are consistent across different populations across sex, ethnicity and aetiology. We identified a common epitope (CE1) shared among 39% of reactive viral antigens that belong to the rhinovirus and enterovirus families. We demonstrated that CE1 could induce both CD4⁺ and CD8⁺ T-cell activation and CD8⁺ T-cell-mediated HCC cell killing.

Conclusions Our results suggest that past exposures to members of the Enterovirus genus may be advantageous for cancer patients, highlighting the potential for a viral peptide-based HCC vaccine.

INTRODUCTION

Hepatocellular carcinoma (HCC) is a heterogeneous malignancy associated with various chronic liver diseases (CLDs), such as hepatitis B and C viruses (HCV), alcohol and dietary-related liver metabolic disorders.^{1,2} Patients with CLD have an increased risk of HCC,³ but not all CLD patients develop cancer. The extent to which host factors may contribute to preventing hepatocarcinogenesis is currently unclear.

Viruses can alter the host immune system and influence carcinogenesis.⁴ While some viruses are pro-carcinogenic, evidence also indicates

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ Viruses are known to alter the host immune system and play a role in the development of hepatocellular carcinoma (HCC) and other malignancies.
- ⇒ Tumour-associated microbial antigens could activate CD8⁺ T cells against melanoma.

WHAT THIS STUDY ADDS

- ⇒ Pan-viral screening reveals that patients with HCC exhibit reduced reactivity to a specific set of viral antigens compared with healthy individuals.
- ⇒ Reactivity to this unique set of viral antigens predicts clinical outcomes associated with HCC.
- ⇒ A common epitope (CE1) is shared among these beneficial viral antigens.
- ⇒ CE1 induces antigen-specific cytolytic activity of CD8⁺ T cells against HCC cells.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ The result of the current study supports the potential of a viral peptide-based cancer vaccine strategy.

that viruses may have beneficial properties for their hosts.^{5–6} Epidemiological studies noted spontaneous regression in leukaemia with acute concurrent viral infections,^{7,8} and the potential of using oncolytic viruses for tumour therapy is being actively investigated.⁹ While the mechanisms mediating the beneficial infections to the host are likely complex, one plausible model is that viral antigens stimulate immune responses that target cancer cells. Studies showed that peptides from tumour-associated bacteria can elicit antitumour effects in melanoma.^{10,11} The potential beneficial effects of viral components on the immune system could open new avenues for cancer prevention and therapy.

By profiling viral infection history among healthy individuals and HCC patients using a synthetic human virome library,¹² we recently identified a viral exposure signature that can identify HCC prior to a clinical diagnosis.¹³ Notably, viral species in this signature showed

differential enrichment, such as HCV being preferentially enriched in HCC, consistent with their pro-carcinogenic roles,¹⁴ while many other viruses are preferentially depleted in HCC. Similar patterns were observed in an independent cohort.¹⁵ We hypothesised that serologically reactive viral antigens enriched in healthy individuals may have protective properties against HCC. To comprehensively search for potentially protective viral antigens, we aimed to determine the serological levels and activities of reactive viral epitope-specific antibodies that are significantly higher in healthy individuals compared with HCC patients across 3 independent cohorts totalling 2647 individuals. We identified a shared viral antigen belonging to the Enterovirus (EV) genus that exhibits strong B-cell activity and predictive T-cell activity. In this study, we tested whether the shared viral antigen is biologically active in inducing T-cells and T-cell-mediated HCC cell killing.

MATERIALS AND METHODS

Materials and methods are included in the online supplemental materials.

RESULTS

Viral epitope-specific antibody repertoires linked to HCC

We accessed serological antibody repertoire data corresponding to 93 904 viral peptides, representing 206 human viral species and over 1000 viral strains available in the National Cancer Institute - University of Maryland (NCI-UMD) cohort¹³ (table 1 and online supplemental table 1). Peptide counts were normalised to antibody epitope binding signal (EBS).¹⁶ Among the 377 unique viral peptides with differential enrichment (online supplemental table 2), 153 viral peptides had significantly higher EBS levels in healthy population control (PC, n=412) than HCC patients (n=148) (figure 1A and online supplemental figure 1A).

Table 1 Demographic and clinical characteristics of the study cohorts*

Variable	NCI-UMD cohort (N=897)	TIGER-LC cohort (N=1577)	NIDDK cohort (N=173)
Age—year			
Median	60	55	52
Range	19–87	21–80	23–79
Sex—no. (%)			
Female	169 (18.8)	365 (23.1)	54 (31.2)
Male	728 (81.2)	1212 (76.9)	119 (68.8)
Race—no. (%)			
Caucasian	347 (38.7)	0	85 (49.1)
African American	533 (59.4)	0	41 (23.7)
Asian	8 (0.9)	1485 (94.2)	34 (19.7)
Other	2 (0.2)	0	2 (1.2)
Missing data	7 (0.8)	92 (5.8)	11 (6.4)
HCV—no. (%)			
Negative	79 (8.8)	1289 (81.7)	59 (34.1)
Positive	360 (40.1)	185 (11.7)	114 (65.9)
Missing data	458 (51.1)	103 (6.5)	0
Hepatitis B virus—no. (%)			
Negative	395 (44.0)	1017 (64.5)	136 (78.6)
Positive	34 (3.8)	485 (30.8)	37 (21.4)
Missing data	468 (52.2)	75 (4.8)	0
Cirrhosis—no. (%)			
Negative	200 (22.3)	0	123 (71.1)
Positive	243 (27.1)	0	44 (25.4)
Missing data	454 (50.6)	1577 (100)	6 (3.5)
Alanine aminotransferase—no. (%)			
Normal, ≤50 U/L	261 (29.1)	1179 (74.8)	51 (29.5)
Elevated, >50 U/L	165 (18.4)	389 (24.7)	122 (70.5)
Missing data	471 (52.5)	9 (0.6)	0
Alpha-fetoprotein—no. (%)			
Negative, ≤20 ng/mL	133 (14.8)	1082 (68.6)	151 (87.3)
Positive, >20 ng/mL	53 (5.9)	468 (29.7)	21 (12.1)
Missing data	711 (79.3)	27 (1.7)	1 (0.6)
Liver disease—no. (%)			
Negative	412 (45.9)	665 (42.2)	0
Chronic liver disease	337 (37.6)	189 (12.0)	129 (74.6)
HCC	148 (16.5)	723 (45.8)	44 (25.4)

*The breakdown clinical characteristics of the cohorts are listed in online supplemental table 1. The hepatitis C carrier was defined based on anti-HCV test. The hepatitis B virus was defined based on HBsAg status.
HBsAg, Hepatitis B surface antigen; HCC, hepatocellular carcinoma; HCV, hepatitis C virus; NCI-UMD, National Cancer Institute - University of Maryland; NIDDK, National Institute of Diabetes and Digestive and Kidney Diseases; TIGER-LC, Thailand Initiative in Genomics and Expression Research for Liver Cancer.

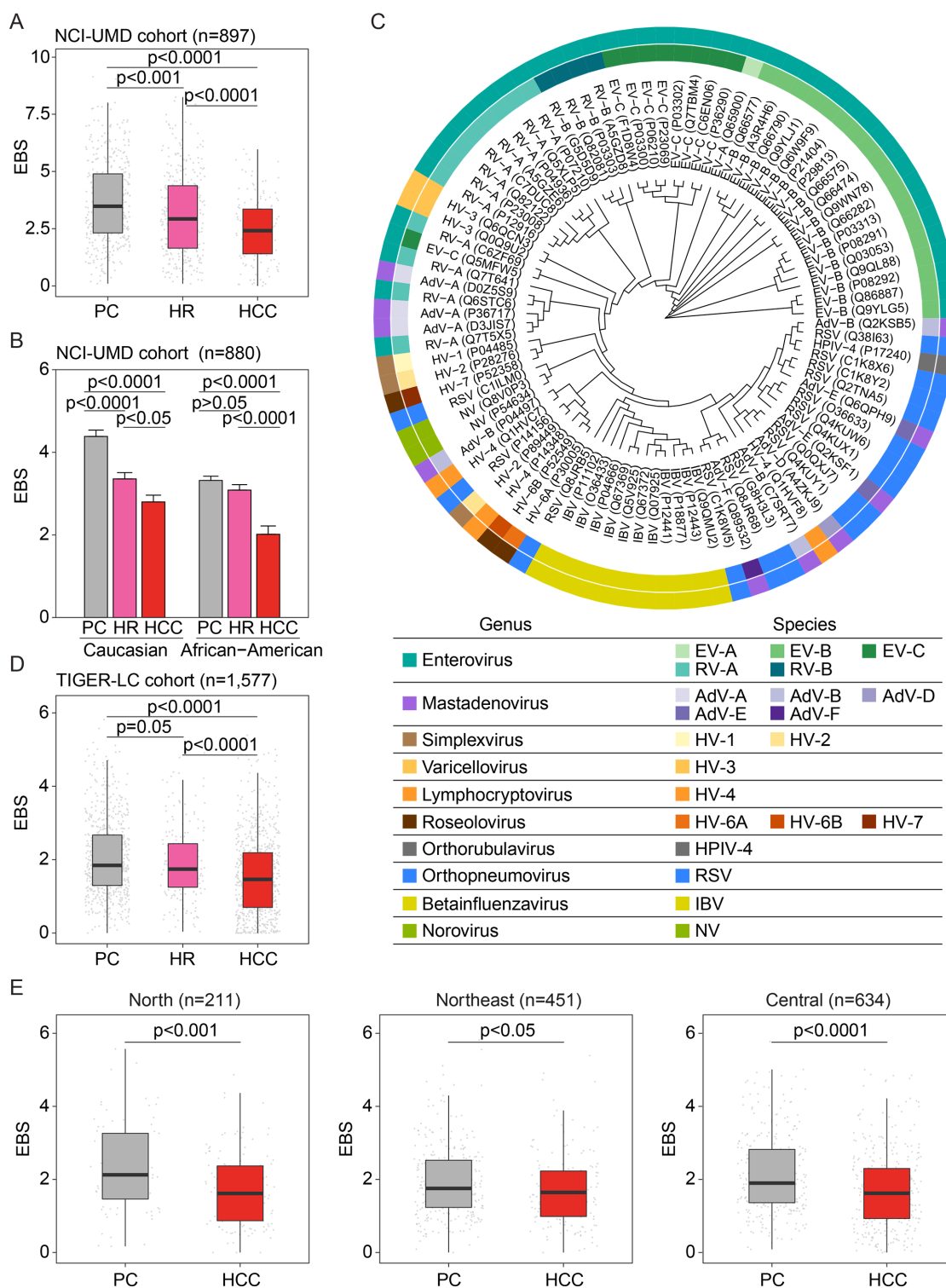


Figure 1 Diminished reactivities of a unique viral epitope-specific antibody repertoire associated with HCC. (A) Average EBS levels of 153 viral peptides in healthy volunteers (PC, n=412), CLD patients (HR, n=337) and patients with HCC (HCC, n=148) in the NCI-UMD cohort. (B) Average EBS levels of 153 viral peptides in PC, HR and HCC stratified by race/ethnicities in the NCI-UMD cohort. Only individuals with a race/ethnicity of Caucasian or African American were included. (C) Phylogenetic tree of 94 viral strains corresponding to 153 viral peptides. Colours in the circles indicate different virus classifications at the species (inner circle) and genus (outer circle) level. (D) Average EBS levels of 153 viral peptides in PC (n=665), HR (n=189) and HCC (n=723) in the TIGER-LC cohort. (E) Average EBS levels of 153 viral peptides in PC and HCC in the TIGER-LC cohort stratified by geographic locations, that is, north, northeast and central regions in Thailand. Individuals with missing geographic information were not included in these plots. In (A, D, E), box plots were applied. The central rectangles span from the first quartile to the third quartile, with lines inside the rectangles indicating the median. Whiskers extend 1.5 times the IQR. Student's t-test was applied to test the statistical significance between different groups. AdV, adenovirus; CLD, chronic liver disease; EBS, epitope binding signal; EV, enterovirus; HCC, hepatocellular carcinoma; HPIV, human parainfluenza virus; HR, high-risk; HV, herpesvirus; IBV, influenza B virus; NCI-UMD, National Cancer Institute - University of Maryland; NV, Norwalk virus; PC, population control; RSV, human respiratory syncytial virus; RV, rhinovirus; TIGER-LC, Thailand Initiative in Genomics and Expression Research for Liver Cancer.

Noticeably, the EBS of the 153 peptides in high-risk individuals (HR, $n=337$) is in between PC and HCC (figure 1A). A similar trend was observed when individuals were stratified by race/ethnicity, but the response was generally higher among Caucasians ($n=347$) than African Americans ($n=533$) (figure 1B). The response was also generally higher among women than men, and among younger persons than older persons, but the differences were not statistically significant (online supplemental figure 1B). This feature contrasted with HCV antigens that were significantly enriched in HCC (online supplemental table 2).¹³ Considering HCV is well characterised for driving HCC, we referred viral strains linked to the 153 PC-enriched antigens as beneficial virus-related antigens. The 153 viral peptides belong to 94 unique proteins across 21 viral species, mainly belonging to the genera of Enterovirus, Mastadenovirus, Orthopneumovirus and Betainfluenzavirus (figure 1C and online supplemental figure 1A). We validated this finding in the Thailand Initiative in Genomics and Expression Research for Liver Cancer (TIGER-LC) cohort, comprising age, sex and locality matched HCC ($n=723$), HR ($n=189$) and PC ($n=665$) (table 1 and online supplemental table 1).¹⁵ Consistently, the EBS of these 153 antigens was higher in the PC than in the HR group ($p=0.05$) and HCC patients ($p<0.0001$) (figure 1D). In addition, this association was stable when individuals were stratified by regions in Thailand (figure 1E).

To determine whether the reduced EBS in HR or HCC was caused by impaired B-cell activities due to underlying liver diseases, we examined serum levels of immunoglobulins, including total IgG, IgA and IgG₄, of participants in the NCI-UMD cohort. In contrast to the decreasing EBS of 153 epitopes in HCC, we found that levels of total IgG, IgG₄ and IgA are significantly higher in HCC than in PC (online supplemental figure 2A). We further examined the interaction between viral repertoire and immunoglobulins and found that the EBS of the 153 peptides consistently showed elevation in PC than HCC, regardless of their immunoglobulin levels, while there were no differences in the EBS of all detectable viral epitopes among HCC and PC (online supplemental figure 2B,C). These findings suggest that the observed reduction in antibody responses to the 153 identified viral epitopes is not due to a general reduction in B-cell activity or antibody levels in HCC. Instead, it appears to be specifically associated with HCC. We also observed similar EBS levels for these epitopes in both cirrhotic and non-cirrhotic individuals within the HR group of the NCI-UMD cohort (online supplemental figure 3A), as well as in HCC patients with and without liver cirrhosis (online supplemental figure 3B). These results further indicate that the 153 identified viral epitopes are not linked to general immune dysregulation, such as that associated with liver cirrhosis, but are specific to HCC. Collectively, a unique set of viral antigens was unveiled with a serological response that was significantly higher in healthy individuals than patients with HCC.

The magnitude and diversity of the beneficial viral antigen-specific antibody repertoires

We performed logistic regression analysis to determine the degree of the enrichment of the 153 antigens in protecting individuals from developing HCC. Based on the quintiles of the EBS levels in PC, we split samples from the PC and HCC groups in the NCI-UMD cohort into five subsets, that is, Q0, Q1, Q2, Q3, Q4, where Q0 had the lowest level of EBS while Q4 had the highest level of EBS. Using Q0 as a reference for logistic regression adjusted by age, sex and race/ethnicity, we found that

the risk of HCC was significantly reduced in Q1, Q2, Q3 and Q4, with an odds ratio (OR) ranging from 0.54 (95% CI 0.30 to 0.98) to 0.06 (95% CI 0.02 to 0.15) (figure 2A). Remarkably, a significant trend of the OR across the five subsets was observed, suggesting a dose-response relationship between the levels of the 153 viral antigen-specific antibodies and risk of HCC (figure 2A).

Hierarchical clustering analysis revealed a relationship of the prevalence of the 153 viral antigens across the five subsets in PC (figure 2B). Three main groups of viral antigens, that is, GR-1, GR-2 and GR-3, were observed. GR-1 comprised members of the rhinovirus (RV), EV, human respiratory syncytial virus, human herpesvirus and influenza B virus, while GR-2 and GR-3 consisted of only RV and EV. Noticeably, GR-2 and GR-3 were enriched in Q4, which had the lowest OR. Consistent findings were observed in the TIGER-LC cohort with the lowest OR in Q4 (figure 2C), indicating that this feature was stable across different geographical locations and race/ethnicities. Noticeably, the trend in the OR was weaker compared with the NCI-UMD cohort, possibly due to population-specific characteristics.

To further assess the robustness of the protective effects of the 153 viral antigens in HCC, we validated our findings in the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) cohort, which has CLD patients with a follow-up period of up to 20 years¹³ (table 1 and online supplemental table 1). Among 173 CLD patients, 44 patients developed HCC and, on average, their EBS of the 153 viral antigens were much lower than HCC-free CLD patients (figure 2D). A similar trend was observed when stratifying patients by race, but the results were not statistically significant, possibly due to the small sample size in this analysis (online supplemental figure 4). Consistent with the observation from the NCI-UMD cohort, the EBS were generally lower in African American patients than Caucasian patients (figure 2E). Notably, CLD patients with a high EBS had a significantly lower risk of HCC with an HR of 0.56 (log rank $p=0.05$) compared with low EBS group (figure 2F).

We also examined the survival significance of the 153 viral antigens and found that the overall survival of unresected HCC patients was significantly better in individuals with high EBS than those with low EBS in both the NCI-UMD cohort (figure 2G) and TIGER-LC cohort (figure 2H). Consistently, this trend was observed in resected HCC patients from the TIGER-LC cohort (online supplemental figure 5A). EBS-related survival was independent of sex, viral hepatitis status, and alpha-fetoprotein in all cohorts but was linked to alanine transaminase and race/ethnicity in the UMD cohort (online supplemental figure 6). When stratifying patients by race/ethnicity, we found that Caucasians had slightly better survival outcomes ($p=0.2$) and a significantly lower risk ($p<0.05$) of developing HCC compared with African Americans (online supplemental figure 7). This was consistent with higher EBS levels of the identified peptides observed in Caucasians. These racial differences in host response to the identified antigens may result from a combination of genetic, immunological, environmental and socioeconomic factors. Further studies with larger cohorts are needed to better interpret these race-related findings. Noticeably, the association between levels of the 153 viral antigen-specific antibodies and HCC risk was robust and not confounded by ages, sexes, races/ethnicities or underlying aetiological factors. Taken together, the above results suggest that the identified viral antigen-specific antibodies may be protective for HCC development.

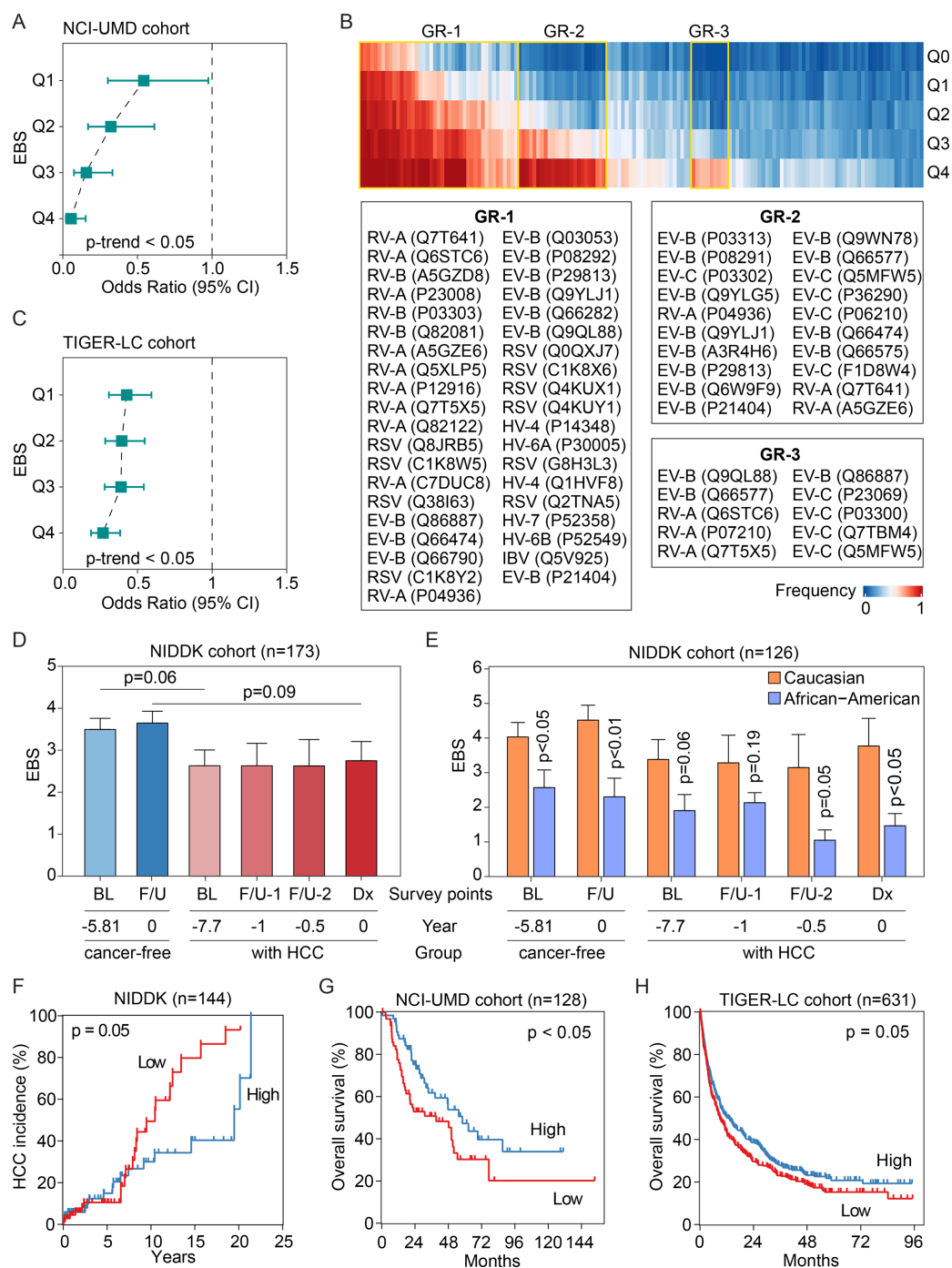


Figure 2 The magnitude and diversity of the beneficial viral epitope-specific antibody repertoire and the degree of its protective effect against HCC. (A) OR derived from logistic regression analysis of PC and HCC in the NCI-UMD cohort. Based on the quintiles of the EBS of the 153 viral peptides in PC, individuals in PC and HCC were partitioned into five subsets, that is, Q0, Q1, Q2, Q3 and Q4. Q0 was used as a reference for the logistic regression analysis, further adjusted by age, sex and race/ethnicity. The dotted line indicates a statistically significant trend of $p < 0.05$. (B) Hierarchical clustering of the 153 reactive viral antigens among 5 equal-sized subsets of PC in the NCI-UMD cohort. GR-1, GR-2 and GR-3 represent three clusters of viral peptides in which the viral protein entrance IDs are indicated in the same order as in the cluster. (C) OR among quintile groups of PC and HCC as Q0, Q1, Q2, Q3 and Q4 in the TIGER-LC cohort. Q0 was used as a reference, further adjusted by age and sex. (D, E) Levels of EBS among CLD patients according to the time of follow-up (D) and stratified by race/ethnicity in the prospective NIDDK cohort (E). Red, CLD patients developed HCC during follow-up; blue, CLD patients were cancer-free during follow-up. Error bar indicates the mean+SEM. (F) Prognostic association of beneficial viral antigens in the NIDDK cohort. CLD patients were stratified into two groups by the median value of the antibody levels to the 153 viral antigens. (G) Overall survival of HCC patients in the NCI-UMD cohort stratified by the median value of the antibody levels to the 153 viral antigens. (H) Overall survival of HCC patients in the TIGER-LC cohort stratified by the median value of the antibody levels to the 153 viral antigens. BL, baseline; CLD, chronic liver disease; Dx, diagnosis; EBS, epitope binding signal; EV, enterovirus; F/U, follow-up; HCC, hepatocellular carcinoma; HV, herpesvirus; IBV, influenza B virus; NCI-UMD, National Cancer Institute - University of Maryland; NIDDK, National Institute of Diabetes and Digestive and Kidney Diseases; PC, population control; RSV, human respiratory syncytial virus; RV, rhinovirus; TIGER-LC, Thailand Initiative in Genomics and Expression Research for Liver Cancer.

Common epitopes shared among viral antigens enriched in healthy individuals

The above results indicate that most of the reactive antigens among the 153 viral peptides belong to RV and EV species, both of which are members of the EV genus. We hypothesised the presence of a common beneficial reactive epitope among them, given their shared characteristic of diminished epitope reactivity in patients with HCC. Considering that the length of each viral peptide in the phage-immunoprecipitation-seq library is 56 residues¹² and that the typical length of an epitope recognised by B and T cells ranges from 8 to 20 amino acids,^{17,18} we searched for consensus sequences among the 153 viral peptides using the Epitope Cluster analysis method with a cluster-break algorithm and a minimum sequence identity threshold >20%, based on the Immune Epitope Database (IEDB).^{19,20} This analysis resulted in 26 subclusters (online supplemental table 3). The largest subcluster contained 60 peptides derived from 40 unique viral strains sharing a sequence similarity, all of which belonged to the EV genus. In contrast, other subclusters contained 1–17 peptides derived from only a few viral strains for each individual cluster (online supplemental table 3). These results indicate that the subcluster with the highest number of viral peptides may have shared anti-HCC activity.

We compared the B-cell epitope score predicted by the IEDB BepiPred Linear Epitope Prediction method to the corresponding EBS values of a representative RV (RV-A, Genbank Accession Q5XLP5) among the 40 unique viral strains. We found a remarkable agreement in epitope responses between the prediction and the patient-derived data (figure 3A). The strongest epitope response spans from 533 to 644 residues of the VP1 region, one of the four viral capsid proteins mediating cell surface attachment through the engagement of different cell surface receptors.²¹ A multiple protein sequence alignment by COBALT revealed a consensus 16-mer peptide (5'-SESVPAL-TAVETGHTS-3', named as CE1), that corresponds to the strongest EBS peak shared by all 40 reactive viral peptides (figure 3B). The EBS levels of these beneficial viral peptides varied among individuals and often showed heterogeneous patterns (figure 3C) and generally had significantly higher levels in PC than in HCC patients (figure 3D).

Biological activities of synthetic peptides derived from RV and EV

We compared the biological activities of a synthetic CE1 peptide (A1-55075), as well as a representative 16-mer native VP1 epitope of RV-B (A2-55087), which had a strong EBS in patients, in peripheral blood mononuclear cells (PBMC) from 10 healthy donors from the NIH blood bank and 8 HCC patients from the NCI-UMD cohort with available viral peptide EBS data and cryopreserved PBMCs (online supplemental table 4). We included an additional native 16-mer peptide, K1654-1, corresponding to the POLG-P3 region of RV-A without any differential reactive EBS among PC and HCC as controls to determine its specificity (online supplemental table 5). We found that A1-55075 and A2-55087 induced cytolytic activities of PMBCs from both healthy donors and HCC patients, while K1654-1 showed no activity (figure 4A and online supplemental figure 8A,B). We noticed that CE1 had a stronger activity than A2-55087 while their activities varied among PBMCs from 18 different individuals. Additionally, we tested a native HCV peptide, K1654-2, having a strong EBS among HCV⁺ HCC patients and found no activity in PBMCs obtained from healthy donors (online supplemental figure 8C).

Next, we measured the EBS levels of EV/RV peptides detected in figure 3B among the individuals involved in the peptide stimulation studies. Consistent with findings from other cohorts, all 10 healthy donors showed prior exposure to EV/RV, with varying EBS levels. We then categorised healthy donors and HCC patients based on the mean EV/RV EBS and compared their PBMC responses to viral peptides. As shown in figure 4B, healthy donors and HCC patients with high EV/RV EBS levels exhibited stronger responses to CE1 and A2-55087 stimulation compared with individuals with low EV/RV EBS levels. This effect was minimal when associated with K1654-1 stimulation. A significant positive correlation was observed between EV/RV EBS levels and responses to CE1 stimulation in 18 PBMCs (figure 4C). Consistently, higher EV/RV EBS in 8 HCC patients was also linked to better overall survival (online supplemental figure 5B). Collectively, these results suggest that serological reactivity to EV/RV peptides correlates with cytolytic responses to ex vivo EV/RV peptides stimulation.

Using intracellular staining, we found that CE1 could induce cytolytic activity in CD4⁺ and CD8⁺ T cells with a similar trend observed in cells treated with A2-55087 (online supplemental figure 8D). These activities aligned with the presence of predicted T-cell and B-cell binding sites in this region (online supplemental figure 9). While substantial evidence supports a coordinated interaction between CD4⁺ T cells and B cells in the adaptive immune response, leading to antibody production and immune memory, HLA class I molecules are crucial for presenting viral peptides directly to CD8⁺ T cells, thereby initiating their cytolytic activity.²² Therefore, we investigated the role of HLA class I in the CE1-associated effects. We performed HLA typing and HLA-peptide binding predictions among the 18 individuals studied (online supplemental table 6). Interestingly, we found that all 18 individuals possessed multiple HLA class I alleles predicted to bind with CE1 and EV/RV reactive peptides (online supplemental table 7).

To identify the region with optimal HLA-recognition, we used T2 cells,²³ a human lymphoblastoid cell line deficient in transporter associated with antigen processing (TAP) and expressing endogenous HLA-A*0201, one of the most common HLA alleles in our PBMC cohort. Due to their TAP deficiency, T2 cells cannot process and present endogenous antigens, allowing only exogenously added peptides to be displayed on HLA molecules. We synthesised seven overlapping 9-mer peptides spanning the native VP-1 (P36290) sequence, covering the entire CE1 region with 3 amino acid overlaps (online supplemental figure 10A). These peptides were incubated with T2 cells along with a fluorescently labelled HLA-A*0201-restricted peptide (ILKEPVHGV²⁴). Flow cytometry analysis revealed that the 9-mer peptide from the central region of CE1, ALTAVETGV, showed the strongest inhibition of the known HLA-A*0201-specific antigen binding to T2 cells, indicating it as the optimal epitope for CE1 (online supplemental figure 10B). Notably, this result is consistent with epitope predictions generated by NetMHCpan v4.1 (online supplemental figure 9). Using HLA multimer staining, we observed that both CE1 and A2-55087 could be presented by HLA class I molecules, as evidenced by an increased CD8⁺ T cells positive for CE1-specific multimers in CE1-treated PBMCs from both healthy individuals and HCC patients (figure 4D and online supplemental figure 8E). Collectively, these results suggest that CE1, and to a lesser extent its native peptide (A2-55087), may induce cytolytic activities in CD8⁺ T cells through antigen-specific presentation by HLA class I molecules.

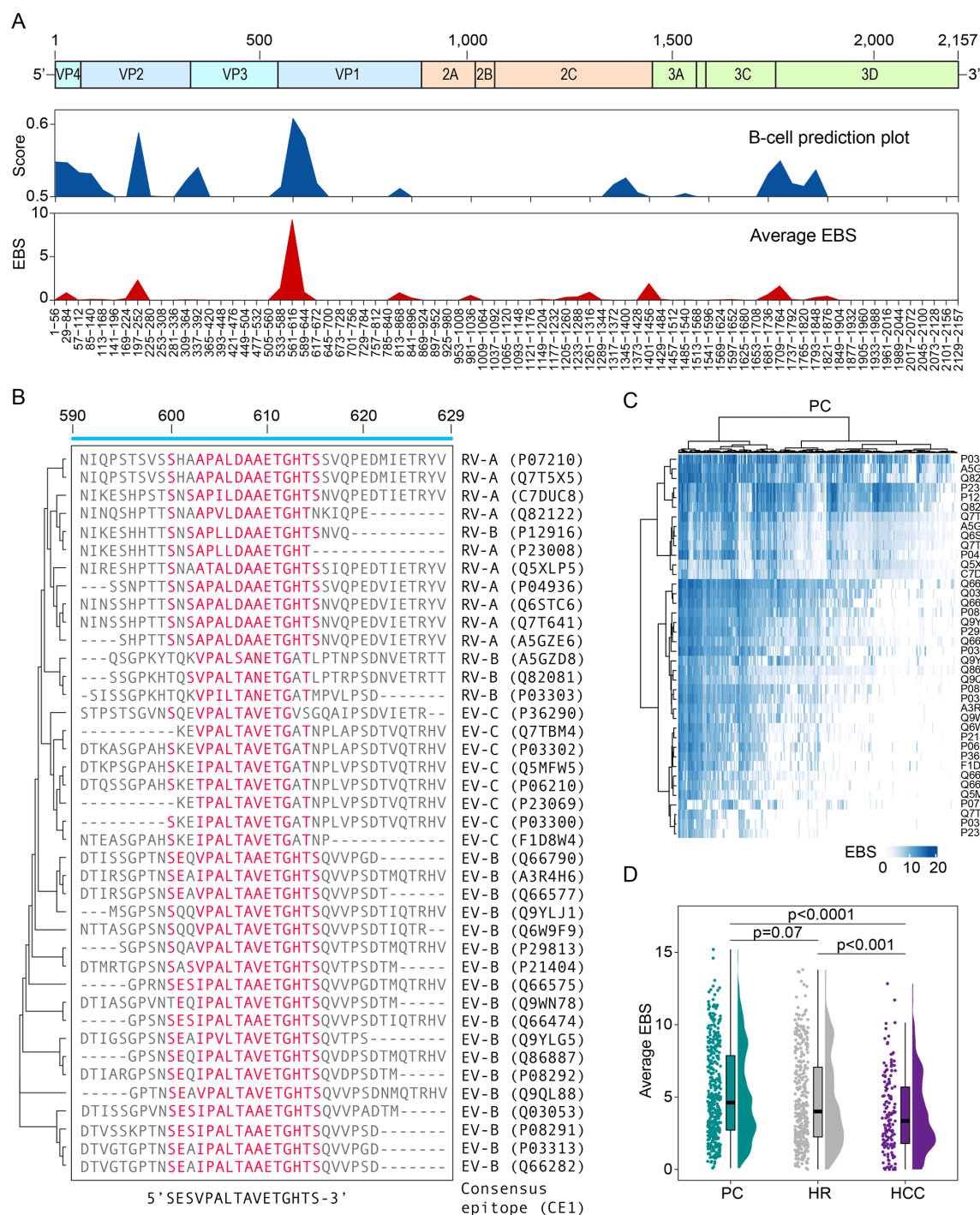


Figure 3 A common epitope (CE1) shared among members of the beneficial viruses. (A) The antigen-specific response of a representative RV (Genbank Accession Q5XLP5) among all subjects in the NCI-UMD cohort and its corresponding B-cell epitope prediction scores. (B) Multiple sequence alignment of the enriched viral peptides of 40 unique viral strains derived from Epitope Cluster analysis. Amino acids are highlighted in red to indicate shared identity among peptides from the indicated RV/EV species. A consensus epitope is indicated. (C) Hierarchical clustering of 40 enriched antigens among PC in the NCI-UMD cohort. (D) Average EBS levels of 40 enriched antigens among PC, HR and HCC in the NCI-UMD cohort. EBS, epitope binding signal; EV, enterovirus; HCC, hepatocellular carcinoma; HR, high-risk; NCI-UMD, National Cancer Institute - University of Maryland; PC, population control; RV, rhinovirus.

Anti-HCC property of CE1-activated CD8+ T cells

To investigate the effects of CE1-induced CD8+T cell activation against HCC, we used PBMC6 (as described above) along with two newly collected PBMC samples from healthy donors (PBMC11 and PBMC12), which were available in large quantities. These cells were subjected to weekly stimulations using irradiated autologous PBMCs loaded with target peptides

(figure 5A). During this period, significant lymphocyte proliferation and expansion were observed, as non-proliferative cells underwent cell death (data not shown). Following this expansion, we isolated CD8+ cells from the expanded PBMC cultures stimulated by A1-55075 and K1654-1, resulting in the creation of CE-CD8-6, Ctrl-CD8-6, CE-CD8-11, CE-CD8-12 and Ctrl-CD8-12 populations (figure 5A). After confirming the purity

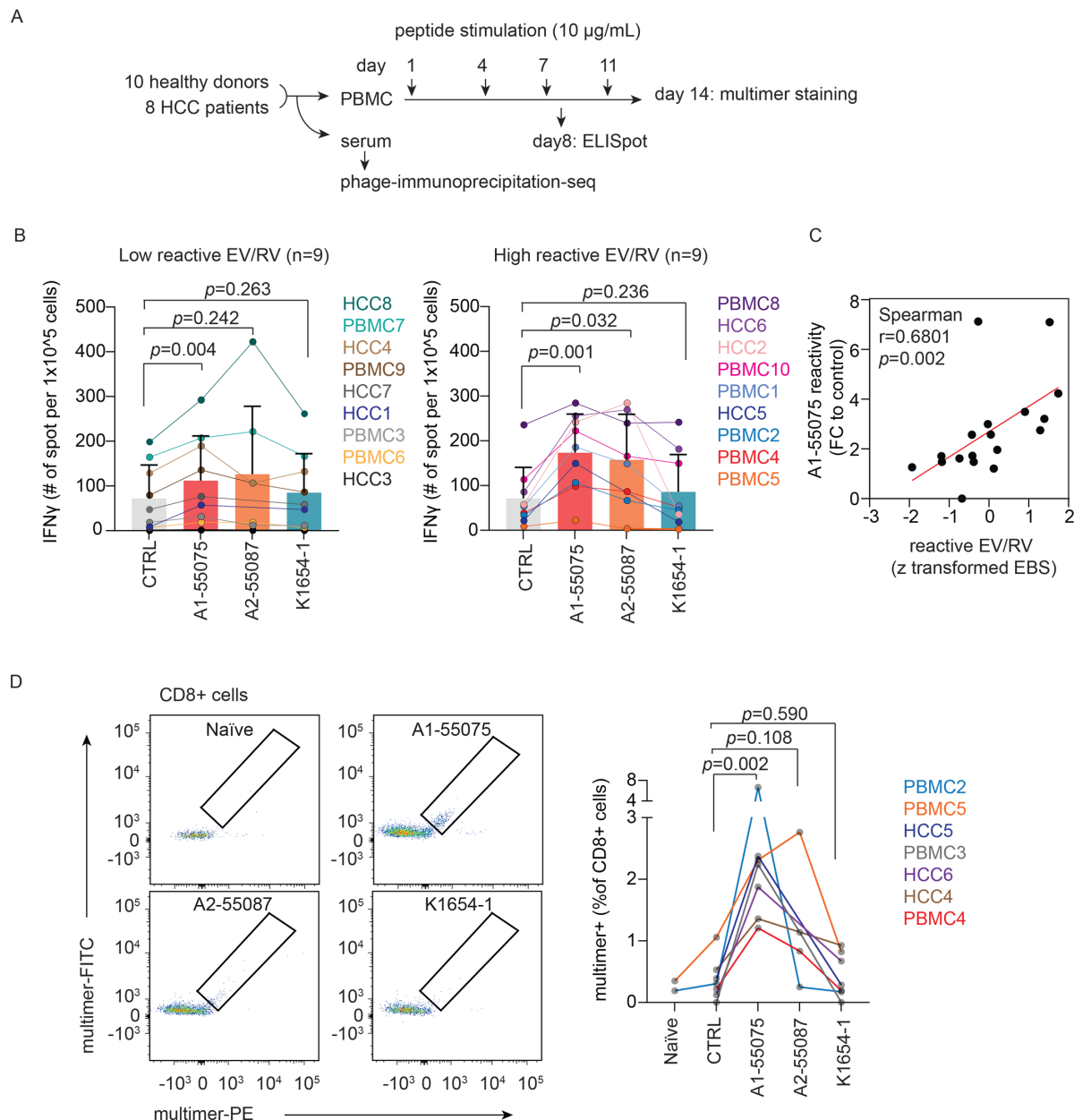


Figure 4 Effects of viral RV/EV-derived peptides on baseline RV/EV serological reactivity in healthy donors and HCC patients. (A) A schematic summary of the testing cohort and experimental procedures (online supplemental table 4). (B) Interferon-gamma (IFN γ) production after peptide stimulation in peripheral blood mononuclear cells (PBMCs) from healthy donors and HCC patients. After stimulation, PBMCs were harvested and assessed for IFN γ production using an ELISpot assay in triplicate. The test cohort was stratified based on the medium z-score transformed EBS of the reactive EV/RV epitopes detected by phage-immunoprecipitation-seq, and the ELISpot results were compared within each subgroup. Dots represent the mean of triplicate ELISpot reads of each testing individual under the indicated conditions. Lines connect the results of individuals across different conditions. Bars summarise the mean across individuals, with error bars representing the SE. Statistical significance was determined using a paired t-test. (C) The relationship between response to ex vivo peptide stimulation and the serological level of reactive EV/RV in healthy donors and HCC patients. The mean EBS levels of EV/RV peptides related to CE1 were z-score transformed to combine the healthy donor cohort (n=10) with HCC patients (n=8). (D) Detection of antigen-specific CD8+T cells in PBMCs after peptide stimulation. PBMCs with HLA-A*0201 or HLA-C*0303 were identified and stained with HLA-A*0201 ALTAVETGV multimers and HLA-C*0303 SANETGATL multimers, respectively. Each multimer was conjugated separately with PE and FITC, and each sample was stained with both formats simultaneously. Multimer expressions were assessed on CD8+ cells, and antigen-specific CD8+T cells were gated on cells with linearly correlated expressions of PE and FITC-multimers. HCV multimers were used as a negative control (online supplemental figure 8E). Representative multimer expressions are shown on the left. The percentage of multimer+CD8+ T cells after peptide stimulation is summarised on the left panel. Dots represent the average results of the multimer staining, and lines connect individuals across different stimulations. Statistical significance was determined using a two-sided independent t-test. For all the experiments, PBMCs were cultured in human IL-2-supplemented medium and stimulated with peptides. EBS, epitope binding signal; EV, enterovirus; FITC, fluorescein isothiocyanate; HCC, hepatocellular carcinoma; HCV, hepatitis C virus; PBMCs, peripheral blood mononuclear cells; PE, phycoerythrin; RV, rhinovirus.

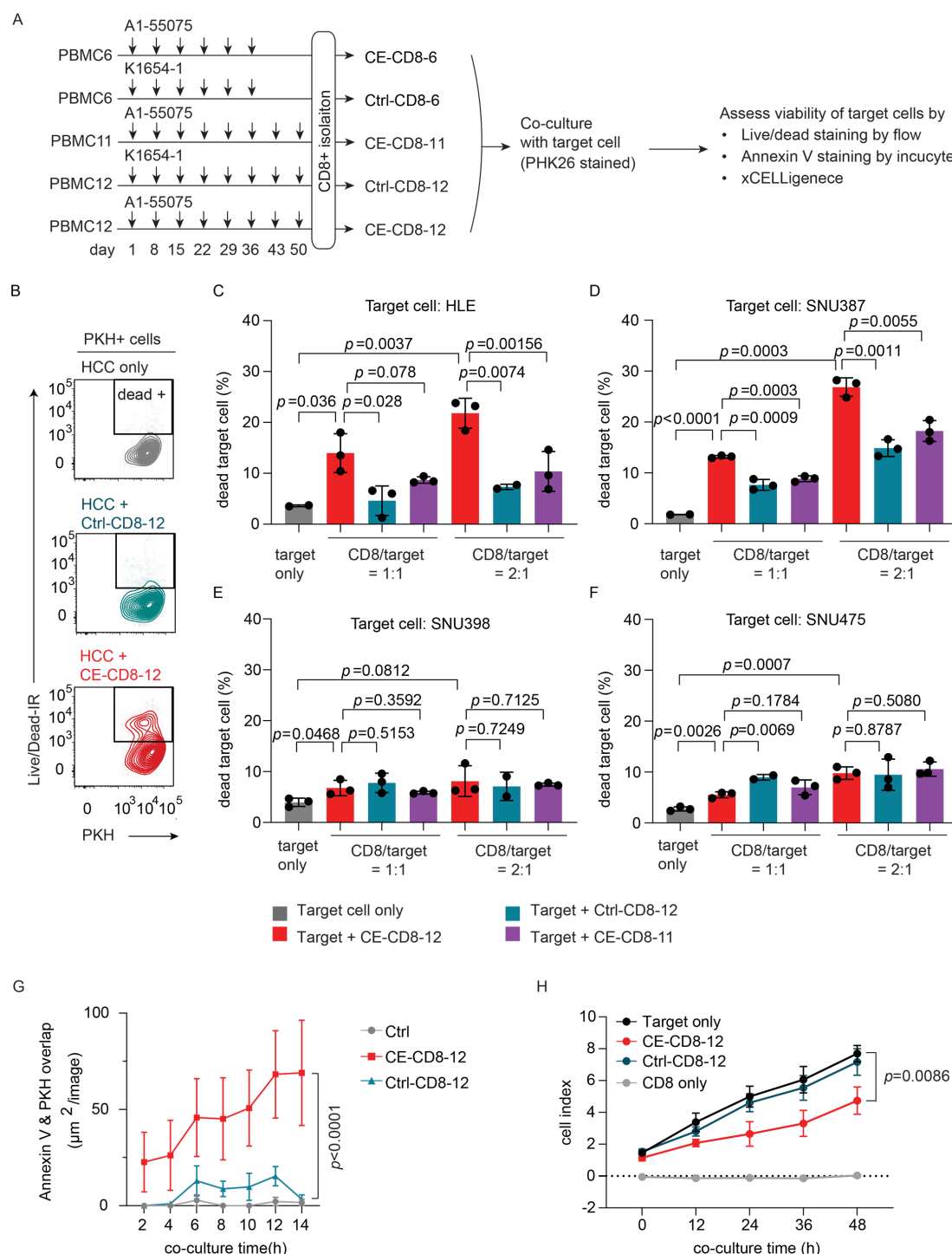


Figure 5 HCC killing effects of peptide-activated T Cells. (A) The diagram summarises the steps of T cell stimulation and co-culture experiments. Briefly, PBMCs from healthy donors, PBMC6, PBMC11 and PBMC12, were cultured in human IL-2-supplemented medium and stimulated weekly with peptides (10 µg/mL) presented by autologous PBMCs for 8 weeks, followed by the isolation of CD8+ T cells (negative selection). Tumour cells were freshly stained with PKH-26, incubated with the indicated CD8+ T cells at a 1:1 or 2:1 T-cell-to-target ratio for 24–48 hours, and then assessed for viability by indicated methods. (B) Representative expression levels of PKH and Live/Dead-IR staining in the HLE and CD8+ T cell mix. The dose-dependent effects of different CD8+ T cells on four HCC cell lines—HLE (C), SNU387 (D), SNU398 (E), and SNU475 (F)—after a 24-hour co-culture were summarised in bar graphs (n=3). Bars represent the mean, and error bars indicate the SD. Statistical significance was determined using a two-sided independent t-test. (G) PKH-stained SNU387 cells were co-cultured with CE-CD8-12 and Ctrl-CD8-12 in the presence of Annexin-V-FITC dye and continuously monitored using Incucyte. The apoptosis of HCC cells was estimated by the overlapping areas of PKH-26 (red) and Annexin-V (green). Dots represent the mean, and error bars indicate SD (n=12). (H) HLE cells were seeded on the E-plate 1 day before co-culture, and the indicated CD8+ T cells were added to the tumour cells at a 2:1 ratio on the second day. The cells were continuously monitored using xCELLigence for 48 hours. Dots represent the mean, and error bars indicate SD. Statistical significance was determined using a two-sided independent t-test. For all co-culture experiments, CD8+ T cells were first suspended in complete medium containing human IL-2 and then added to tumour cells (see online supplemental data for detail). FITC, fluorescein isothiocyanate; HCC, hepatocellular carcinoma; PBMCs, peripheral blood mononuclear cells.

of these isolated CD8+T cells using flow cytometry (online supplemental figure 11A), the cells were either maintained in an antigen-free culture condition or frozen for subsequent peptide stimulation and co-culture experiments.

We assessed antigen specificity by re-exposing these isolated CD8+T cells to their relevant and irrelevant peptides (online supplemental figure 11B). Notably, CE-CD8-11 and CE-CD8-12 exhibited a significant increase in cell proliferation after a 48-hour restimulation with autologous PBMCs loaded with A1-55075. In contrast, no such proliferation was observed when cells were restimulated with scrambled-A1-55075 or K1654-1 peptides (online supplemental figure 11C,D). Ctrl-CD8-12, while showing a higher cell proliferation in response to K1654-1 compared with A1-55075, demonstrated a weaker response overall when compared with the CE-CD8-11 and CE-CD8-12 responses to A1-55075 (online supplemental figure 11E). Collectively, these findings indicate that our method successfully enriched and expanded CD8+T cells with high antigen-specificity.

To determine the HCC killing effects of CE1-specific CD8+T cells, we compared the HLA profile of PBMC6, PBMC11 and PBMC12 with several available HCC cell lines maintained under similar culture conditions as PBMCs (online supplemental table 6). We found that SNU387 and HLE cells share the same HLA allele, C*0303, with PBMC6 and PBMC12, while SNU398 and SNU475 do not. None of the four HCC cell lines share an identical HLA allele with PBMC11. We then incubated these HCC cell lines with the isolated CD8+T cells for 24 hours and assessed the effects on HCC viability using flow cytometry (figure 5A,B and online supplemental figure 11F). As shown in figure 5C,D, CE-CD8-12 induced dose-dependent HCC killing in HLE and SNU387, with significantly greater effects than those observed with Ctrl-CD8-12 and CE-CD8-11. Similar trends were observed in co-culture studies using CE-CD8-6 (online supplemental figure 11G). Interestingly, co-culturing CE-CD8-12 with HLE resulted in increased cytolytic cytokine production compared with CE-CD8-12 alone (online supplemental figure 11H). For SNU398 and SNU475, the tumour killing effects mediated by CE-CD8-12 were significantly reduced, and no differential effects were observed between CE-CD8-12 and Ctrl-CD8-12 and CE-CD8-11 (figure 5E,F).

To further characterise the time-dependent effects of HCC killing mediated by CE-CD8-12 and Ctrl-CD8-12, we applied continuous monitoring. By staining the tumour cells with PKH26 and annexin V, we observed that CE-CD8-12 promoted apoptosis in HCC cells as early as the second hour of co-culture, with a continuously increasing trend, while the effects elicited by Ctrl-CD8-12 occurred later and were of a smaller magnitude (figure 5G and online supplemental figure 11I). Due to the limitation of PKH dye, which faded over time in HCC cells post initial staining, we also employed a stain-free, electrical-impedance-based method to monitor the long-term HCC killing effects. Consistent with our previous observations, HCC cells co-cultured with CE-CD8-12 showed a significant reduction in proliferation compared with those co-culturing with Ctrl-CD8-12 up to 48 hours (figure 5H). Additionally, we examined the effects of CE-CD8-11 and CE-CD8-12 cells on immortalised normal human hepatocyte-derived cell lines HHT3 and HHT4.²⁵ As shown in online supplemental figure 12, neither CE-CD8-11 nor CE-CD8-12 produced significant cytotoxic effects in these normal immortalised human hepatocyte-derived cell lines. Together, these results indicate that CE1-specific CD8+T cells elicit HLA-associated cytotoxicity in HCC cells.

CE1-mediated T-cell clonal expansion

To elucidate the biological activities of CE1, we exposed PBMC12 to weekly CE1 stimulation, as previously described, and performed sequential intracellular staining and multimer staining throughout the peptide stimulation process. CE1 stimulation elicited a time-dependent activation of cytolytic activities in both CD4+and CD8+ T cells, with a marked enrichment of antigen-specific CD8+T cells (figure 6A,B and online supplemental figure 13). To determine whether CE1 induced specific T-cell clonal expansion, we performed single cell studies on PBMC6 and PBMC12 before and after CE1 stimulation (figure 6C). In both donors, we found that the transcriptomic profiles of CE1-activated PBMCs were distinct from those of naïve PBMCs, with T cells showing the most dominant changes after stimulation (figure 6C and online supplemental figure 14A). Notably, CE1 stimulation induced significant functional polarisation in both CD4 and CD8 T cells, where CD8 T effector cells, CD4 T helper cells and T regulatory cells were the most dominant types (figure 6D and online supplemental figure 14B,C). Using single-cell T cell receptor (TCR) sequencing, we demonstrated that CE1 stimulation significantly altered TCR clonality in both CD4 and CD8 T cells, driving expansion of specific clonotypes (figure 6E,F and online supplemental figure 14D,E). In PBMC12, 38.6% (750/1,944) of clonotypes associated with CE1-activated T cells showed expansion (TCR copies ≥ 2 copies), compared with only 2.33% (106/4546) in treatment naïve T cells (figure 6E). A similar pattern was observed in PBMC6 (online supplemental figure 14D). Notably, most of the clonotypes with major expansion (TCR copies ≥ 4) were found in CE1-activated T cells, while naïve T cells predominately contained non-proliferative clonotypes (TCR copy=1). We also observed a few overlapping clonotypes between naïve and CE1-activated T cells, which tended to occur more frequently in CD8 T cells, while the majority of clonotypes were unique to each group (figure 6F and online supplemental figure 14E). Further examination of clonotypes with larger TCR copies among CD4 and CD8 T cells revealed that these clonotypes were predominantly present in CE-activated cells and were associated with CD8 effector cells, CD8 tissue-resident memory T cells, CD4 helper cells and T regulatory cells (figure 6G,H, online supplemental figure 14F,G). Together, these data indicate that CE1 stimulation can effectively activate T cells with terminal differentiation, characterised by unique TCR clonal expansion in PBMCs.

DISCUSSION

By analysing pan-viral serological antibody repertoires, we uncovered a unique set of RV/EV antibodies more abundant in cancer-free individuals than HCC patients. The RV/EV serological activity predicted the risk of HCC development in CLD patients and the mortality risk in HCC patients. This finding was consistent among 2647 US and Thai individuals of different sexes, ethnicities and aetiologies. Noticeably, the host response to RV/EV antigens was much higher in European Americans than African Americans, and much higher in women than men, which coincides with epidemiological observations for differential incidence of liver cancer among race/ethnicity and sex. However, these results need to be further evaluated in large cohort sizes. We identified a common epitope, CE1, with the strongest EBS shared among reactive RV/EV antigens. This epitope exists in a linear form within the viral VP1 sequence, based on available crystallographic structure data. Notably, healthy individuals exhibited significantly higher levels of circulating CE1-related

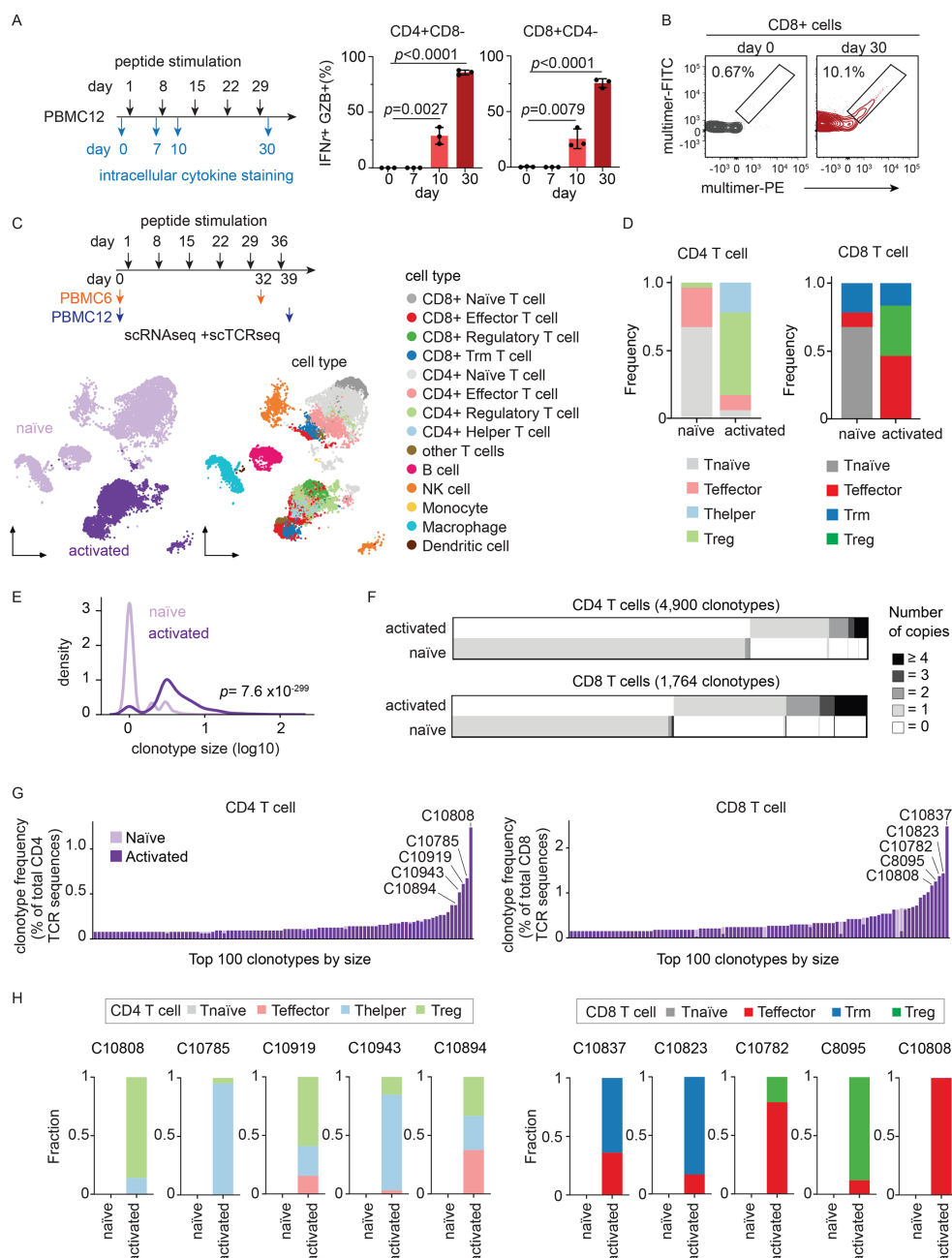


Figure 6 Time-dependent effects of CE1 on PBMCs and T cell clonality. (A) The effects of CE1 on CD4+ and CD8+ T cells over time. PBMCs from a healthy donor (PBMC12) were stimulated with the CE1 peptide (A1-55075, 10 µg/mL or 5.9 µM) on day 1, then restimulated weekly with CE1-loaded autologous antigen-presenting cells for four additional weeks. PBMCs were collected at specified time points, intracellularly stained and analysed by flow cytometry for IFN γ and granzyme B (GZB). Throughout the experiment, PBMCs were continuously maintained in complete medium supplemented with human IL-2 and stimulated with peptides. The diagram on the left summarises the experimental procedure, while the two panels on the right show IFN γ and GZB expression in CD4+ and CD8+ T cells. Bars represent the mean, and error bars indicate the SD. Statistical significance was determined using a two-sided independent t-test. (B) Detection of antigen-specific CD8+ T cells in PBMC12 before and after 5 week CE1 stimulation, using HLA-C*0303 SANETGATL-PE and HLA-C*0303 SANETGATL-FITC. The level of multimers on CD8+ cells is shown. HLA-C*0303 HCV-PE and HLA-C*0303 HCV-FITC were used as negative controls. (C) Characterisation of CE1 effects on PBMCs using single cell analysis. The top diagram summarises the experimental scheme, with UMAP plots below showing all PBMCs associated with PBMC12, coloured by CE1 stimulation status (left) and cell types (right). As described above, PBMCs were continuously maintained in complete medium supplemented with human IL-2 and received weekly peptide stimulation. Treatment-naïve cells were recovered in IL-2 containing medium for 1 day prior to capture (detailed in online supplemental data). (D) Bar graphs summarise the composition of naïve and CE1-activated CD4 and CD8 T cells associated with PBMC12. (E) A histogram shows the distribution of TCR clonotype size in naïve and CE1-activated T cells associated with PBMC12. Anderson-Darling test was used to examine the difference of distribution between two groups. (F) A heatmap summarises the intensity of unique TCR clonotypes in naïve and CE1-activated CD4 and CD8 T cells associated with PBMC12. (G) The top 100 clonotypes among CD4 and CD8 T cells are displayed, with bar size corresponding to clonotype frequency and colours indicating CE1 stimulation status associated with PBMC12. (H) The cell types associated with the top five clonotypes in CD4 and CD8 T cells are shown, with bar size corresponding to frequency and colours indicating cell type associated with PBMC12. PBMCs, peripheral blood mononuclear cells.

antibodies compared with HCC patients, suggesting that regions with high EBS correspond to epitopes recognised by immunoglobulins or B cells. Interesting, the EBS pattern closely matched bioinformatic predictions of B cell epitopes, with the highest scoring region overlapping with CE1. This suggests that CE1 may play a key role in mediating B cell immunity. Experimentally, we demonstrated that a synthetic CE1 peptide is biologically active in inducing T cell cytolytic activity, TCR clonal expansion of both CD4 and CD8 T cells, and HLA-dependent cytotoxicity against HCC cells. Our study meets Hill's criteria for causation.²⁶ First, we performed our investigation in three independent HCC cohorts with matched CLD and healthy controls and consistently identified a diminished RV/EV antibody response associated with HCC occurrence and mortality, supporting a protective role. Second, a prospective cohort analysis with up to 20-year follow-up demonstrates the temporality of this association. Moreover, we observed a dose-dependent protective effect of RV/EV against HCC, supporting evidence of a biological gradient. Finally, the functional properties of the CE1 peptide are experimentally confirmed. We demonstrated that CE1 induces cytolytic activity of both CD4+ and CD8+ T cells from PBMC and HCC cell killing in an HLA-specific manner.

In our study, we demonstrated that CE1, a peptide sequence commonly shared among reactive EV/RV epitopes, activates both CD4+ and CD8+ T cells. The central region of CE1 showed the strongest binding to HLA class I molecules. Similarly, Gaido *et al* used a synthetic overlapping peptide library spanning the entire VP-1 protein of RV-C3 and identified a 15-mer peptide largely overlapping with the CE1 region that most effectively stimulated CD4+ T cell proliferation among all VP-1 fragments.²⁷ Together, the data suggest that the protective effects of EV/RV against HCC may involve both cellular and humoral immune responses. However, the mechanisms by which EV/RV-immunity translates to HCC protection in clinical cases remain unclear. Most RV and EV use intercellular adhesion molecule or low-density lipoprotein receptor for cell entry,²⁸ both of which are ubiquitously expressed in airway epithelial cells and hepatocytes. One possibility is that CE1-containing virions are directly processed by HCC cells, which often express both HLA-I and HLA-II alleles. This is similar to findings from a recent study in melanoma, where tumour cells presented intracellular bacterial-derived peptides and elicited immune responses.¹⁰ To investigate this possibility, we analysed single-cell RNA sequencing data from HCC patients²⁹ for CE1-related viral sequences. However, no sequences matching or resembling CE1 were detected (data not shown). An alternative explanation is that CE1-containing virions are processed and presented by APCs, which in turn activate CE1-specific CD4+ and CD8+ T cells. These T cells may then cross-react with cellular proteins, exerting cytotoxic effects on tumour cells. This model is biologically plausible, as pathogens such as viruses and bacteria have been shown to exploit molecular mimicry to influence host immunity.^{30 31} This mechanism has been implicated in diseases such as multiple sclerosis, where Epstein-Barr virus has been linked to autoimmunity.^{32 33} Our data support this model. HCC cells, even without prior exposure to CE1, responded to CE1-specific CD8+ T cells in an HLA-restricted manner. Although the identity of target cellular proteins remains unknown, BLAST analysis identified several human protein fragments with high similarity to CE1 (online supplemental figure 15). Further studies using immunopeptidomics or related approaches in HCC tumour biospecimens are needed to determine whether the observed immune responses are driven by endogenously expressed CE1-mimicking antigens

or by exogenous viral peptides processed and presented by HCC cells to explain the protective effects of EV and RV.

Our findings suggest the potential of a CE1-based cancer vaccine for HCC prevention. RV and EV generally cause mild clinical symptoms.²⁸ The high prevalence of circulating antibodies against RV/EV in the general population suggests that most adults may have already developed immunological memory against these viruses. We observed a correlation between circulating RV/EV-specific antibody levels and ex vivo CE reactivity, indicating effective immune memory against EV/RV in individuals with high EV/RV EBS regardless of disease status. It is possible that the current in vitro experimental setup does not fully replicate the complex immune environment in HCC patients, which may explain the lack of observable differences between HCC patients and healthy controls. It should be noted that we only tested 18 donors with available cryopreserved PBMCs. To address this limitation, we used two immune-competent mouse models with established syngeneic liver tumours, that is, BALB/c and C57BL/6 mice. We isolated splenocytes from these mice and tested their responses to in vitro CE1 stimulation, the same strategy used in human PBMC studies. Despite multiple attempts, we did not observe CD4+ and CD8+ T cell activation (online supplemental figure 16). Supporting this result, peptide-MHC binding predictions indicated that CE1 does not contain epitopes recognised by MHC alleles of BALB/c and C57BL/6 mice. These findings suggest that CE1 is likely recognised uniquely by the human immune system. Future studies using humanised mouse models or nonhuman primates may help further explore this mechanism.

In the classical model, CD8+ T cell-mediated tumour killing depends on the recognition of antigen-MHC complexes on tumour cells. In our study, CE1-stimulated CD8+ T cells exhibited significantly stronger cytotoxic activity against HLA-matched HCC cells compared with Ctrl-stimulated CD8+ T cells. This effect was not observed in HLA-unmatched HCC cells, indicating that CE1-mediated cytotoxicity is antigen-specific and HLA-specific. Interestingly, we also observed some degree of antigen-independent cytotoxicity in certain HCC cell lines. For example, the SNU387 cell showed increased cell death when co-cultured with both Ctrl-CD8-12 and CE-CD8-11 cells, despite the absence of matched antigen and HLA presentation. Notably, these antigen-independent HCC killing is heterogeneous among HCC cell lines, suggesting that multiple regulatory mechanisms may be involved. Potential contributors include cytokine dysregulation and interactions between CD8+ T cells and tumour cell ligands such as NKG2D.^{34 35} Further studies are warranted to elucidate the mechanisms underlying the heterogeneous sensitivity of HCC cells to immune-mediated killing.

Our results highlight the presence of CE1-associated immune memory in both healthy individuals and HCC patients, suggesting a promising opportunity to leverage this immune memory for anti-HCC strategies. EV/RV EBS levels may serve as a predictive biomarker for immune responsiveness. In parallel, one could also consider the utility in developing a serological response test to CE1 as a predictive biomarker for monitoring the efficacy of a vaccine-based approach. This approach may also complement other treatment modalities, such as surgical resection, to reduce the risk of tumour recurrence. Moreover, since this mechanism may extend beyond HCC, profiling pan-viral serological antibody repertoires across different cancer types represents a logical next step for future investigation.

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Contributors Conceptualisation: LM and XWW; Methodology: LM, MHH and XWW; Investigation: LM, MHH, FRM, LW, QL, MF, KCW, AB, JCa, JCh, SR, BP, MR and XWW; Visualisation: LM, MHH, FRM and XWW; Funding acquisition: XWW; Project administration: XWW; Supervision: XWW; Writing—original draft: LM, MHH and XWW; Writing—review and editing: all authors; XWW is the guarantor.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Consent obtained directly from patient(s).

Ethics approval This study was approved by the Institution Review Boards of the respective institution in Thailand and USA (NCI protocol number 13CN089; CRI protocol number 18/2555; Chulabhorn Hospital protocol number 11/2553; Thai NCI protocol number EC163/2010; Chiang Mai University protocol number TIGER-LC; Khon Kaen University protocol number HE541099). Informed consent was obtained from all participants included in this study.

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Data availability statement Data are available in a public, open access repository. The viral exposure profiles of the TIGER-LC cohort are available at Harvard dataverse (<https://doi.org/10.7910/DVN/NIT39Z>). The viral exposure profiles of the NCI-UMD and the NIDDK cohorts are available from the corresponding author on request. The single-cell data of the PBMCs are available at GEO database (GEO284796). All other data generated in this study are included in this published article and online supplemental materials. This paper does not report original code or generate new unique reagents.

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REFERENCES

- Wang XW, Thorgeirsson SS. The biological and clinical challenge of liver cancer heterogeneity. *Hepat Oncol* 2014;1:349–53.
- Ma L, Wang XW. Dissecting liver tumor heterogeneity to improve health equity. *Trends Cancer* 2022;8:286–90.
- Arzumanyan A, Reis HMGPV, Feitelson MA. Pathogenic mechanisms in HBV- and HCV-associated hepatocellular carcinoma. *Nat Rev Cancer* 2013;13:123–35.
- Cadwell K. The virome in host health and disease. *Immunity* 2015;42:805–13.
- Roossinck MJ, Bazán ER. Symbiosis: Viruses as Intimate Partners. *Annu Rev Virol* 2017;4:123–39.
- Sachs JL, Essenberg CJ, Turcotte MM. New paradigms for the evolution of beneficial infections. *Trends Ecol Evol* 2011;26:202–9.
- PELNER L, FOWLER GA, NAUTS HC. Effects of concurrent infections and their toxins on the course of leukemia. *Acta Med Scand Suppl* 1958;338:1–47.
- Dock G. The influence of complicating diseases upon leukaemia.*. *Am J Med Sci* 1904;127:563–92.
- Bell J, McFadden G. Viruses for tumor therapy. *Cell Host Microbe* 2014;15:260–5.
- Kalaora S, Nagler A, Nejman D, et al. Identification of bacteria-derived HLA-bound peptides in melanoma. *Nature New Biol* 2021;592:138–43.
- Naghavian R, Faigle W, Oldrati P, et al. Microbial peptides activate tumour-infiltrating lymphocytes in glioblastoma. *Nature New Biol* 2023;617:807–17.
- Xu GJ, Kula T, Xu Q, et al. Viral immunology. Comprehensive serological profiling of human populations using a synthetic human virome. *Science* 2015;348:aaa0698.
- Liu J, Tang W, Budhu A, et al. A Viral Exposure Signature Defines Early Onset of Hepatocellular Carcinoma. *Cell* 2020;182:317–28.
- Farazi PA, DePinho RA. Hepatocellular carcinoma pathogenesis: from genes to environment. *Nat Rev Cancer* 2006;6:674–87.
- Do WL, Wang L, Forgues M, et al. Pan-viral serology uncovers distinct virome patterns as risk predictors of hepatocellular carcinoma and intrahepatic cholangiocarcinoma. *Cell Rep Med* 2023;4:101328.
- Mina MJ, Kula T, Leng Y, et al. Measles virus infection diminishes preexisting antibodies that offer protection from other pathogens. *Science* 2019;366:599–606.
- Singh H, Ansari HR, Raghava GPS. Improved method for linear B-cell epitope prediction using antigen's primary sequence. *PLoS One* 2013;8:e62216.
- Reynisson B, Alvarez B, Paul S, et al. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. *Nucleic Acids Res* 2020;48:W449–54.
- Vita R, Mahajan S, Overton JA, et al. The Immune Epitope Database (IEDB): 2018 update. *Nucleic Acids Res* 2019;47:D339–43.
- Dhanda SK, Vaughan K, Schulten V, et al. Development of a novel clustering tool for linear peptide sequences. *Immunology* 2018;155:331–45.
- Palmenberg AC, Gern JE. Classification and evolution of human rhinoviruses. *Methods Mol Biol* 2015;1221:1–10.
- Rock KL, Reits E, Neefjes J. Present Yourself! By MHC Class I and MHC Class II Molecules. *Trends Immunol* 2016;37:724–37.
- Day PM, Esquivel F, Lukso J, et al. Effect of TAP on the generation and intracellular trafficking of peptide-receptive major histocompatibility complex class I molecules. *Immunity* 1995;2:137–47.
- Brander C, Hartman KE, Trocha AK, et al. Lack of strong immune selection pressure by the immunodominant, HLA-A*0201-restricted cytotoxic T lymphocyte response in chronic human immunodeficiency virus-1 infection. *J Clin Invest* 1998;101:2559–66.
- Jiang W, Wang XW, Unger T, et al. Cooperation of tumor-derived HBx mutants and p53-249(ser) mutant in regulating cell proliferation, anchorage-independent growth and aneuploidy in a telomerase-immortalized normal human hepatocyte-derived cell line. *Int J Cancer* 2010;127:1011–20.
- HILL AB. THE ENVIRONMENT AND DISEASE: ASSOCIATION OR CAUSATION? *Proc R Soc Med* 1965;58:295–300.
- Gaido CM, Stone S, Chopra A, et al. Immunodominant T-Cell Epitopes in the VP1 Capsid Protein of Rhinovirus Species A and C. *J Virol* 2016;90:10459–71.
- Royston L, Tapparel C. Rhinoviruses and Respiratory Enteroviruses: Not as Simple as ABC. *Viruses* 2016;8:16.
- Ma L, Heinrich S, Wang L, et al. Multiregional single-cell dissection of tumor and immune cells reveals stable lock-and-key features in liver cancer. *Nat Commun* 2022;13:7533.
- Alcami A. Viral mimicry of cytokines, chemokines and their receptors. *Nat Rev Immunol* 2003;3:36–50.
- Elde NC, Malik HS. The evolutionary conundrum of pathogen mimicry. *Nat Rev Microbiol* 2009;7:787–97.
- Lanz TV, Brewer RC, Ho PP, et al. Clonally expanded B cells in multiple sclerosis bind EBV EBNA1 and GlialCAM. *Nature New Biol* 2022;603:321–7.
- Tengvall K, Huang J, Hellström C, et al. Molecular mimicry between Ancylostoma 2 and Epstein-Barr virus nuclear antigen 1 associates with multiple sclerosis risk. *Proc Natl Acad Sci U S A* 2019;116:16955–60.
- Lerner EC, Woroniecka KI, D'Anniballe VM, et al. CD8⁺ T cells maintain killing of MHC-I-negative tumor cells through the NKG2D-NKG2DL axis. *Nat Cancer* 2023;4:1258–72.
- Freeman BE, Hammarlund E, Raué H-P, et al. Regulation of innate CD8⁺ T-cell activation mediated by cytokines. *Proc Natl Acad Sci U S A* 2012;109:9971–6.